

Green Lab - LAB 3

R with Positron

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0. Setup

0.1: Download [Positron](#) and create a new folder for this lab

0.2: Install and load the following packages:

- *tidyverse*
- *ggplot2*
- *bestNormalize*
- *ARTool*
- *lmerTest*
- *xtable*

0.3: In a directory [data/Run_Table_TPCH.csv](#) there is data describing results of Data Analysis Tasks Benchmarking.

1. Data Loading and Inspection

1.1: Load the dataset `Run_Table_TPCCH.csv` into R.

1.2: Keep only the four required columns: *library*, *dataframe_size*, *trial*, and *energy_usage*.

1.3: Convert *library*, *dataframe_size*, and *trial* into factor variables.

1.4: Inspect the structure of the dataset with functions such as `glimpse()`, `summary()`, and `head()` to confirm that the data is properly loaded and formatted.

2. Visual Exploration via Plotting

2.1: Create a histogram of *energy_usage* to inspect its overall distribution and assess whether it appears normal.

2.2: Create boxplots of *energy_usage* grouped by *library* and *dataframe_size* to compare distributions across experimental conditions.

2.3: Create scatterplots of *energy_usage* across *trials*. Assign different colors to the combinations of *library* and *dataframe_size* (i.e., the “experiments”), and include a legend to distinguish them.

3. Data Normality and Normalization

- 3.1:** For each combination of *library* and *dataframe_size*, run a normality test (Shapiro–Wilk).
- 3.2:** Normalize the *energy_usage* column using the **bestNormalize** package and create a new normalized column.
- 3.3:** Plot the normalized data and repeat normality tests.
- 3.4:** Evaluate whether parametric tests can be applied after normalization.

4. (Non-)Parametric Tests

4.1: Use a non-parametric test *wilcox.test* (Wilcoxon's test) to check whether energy usage is different between libraries for small dataframes.

4.2: Fit an *art()* model to observe influence of *library*, *dataframe_size* and its interactions with *energy_used*. Treat trial as a random effect for repeated measures experiment. What can you see? Validate your model with *summary(model)*.

4.3: Similarly to *art*, run *lmer()* model to observe similar effects, except your target variable should be *norm_energy_used*.

4.4: Check residuals for normality and assess the validity of the model.

5. Summary Plots

- 5.1:** Create a boxplot to visualize the distribution of *energy_usage* for each “experiment”. Use *facet_wrap()* to adequately separate the experiments by library or dataframe size.
- 5.2:** Label the x-axis as “Dataframe Size” and the y-axis as “Energy Usage”
- 5.3:** Create a scatter plot (*geom_point()*) to show how *energy_usage* varies over trial. Assign a unique color to each “experiment” and add a legend.
- 5.4:** Using faceting, create a grid with 4 scatter plots (one per “experiment”) showing how *energy_usage* correlates to *execution_time*, Including trend lines.

6. Annotated Tables

6.1: Use **dplyr** functions to calculate the **average** *cpu_usage*, *memory_usage* and *energy_usage* for each “experiment”.

6.2: Using the **xtable** package, format the summary table as LaTeX. You can copy the output for rendering in engines such as Overleaf.